MMDU MULLANA OFFICIAL SYLLABUS • BCSE-004 • UNIT 2

Unit 2: Decision Making, Branching and Functions (Modularity & Scope)

Subject: Python Programming (BCSE-004) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 30 mins

*Control structures (if-elif-else, truthy vs falsy values), while and for loops, range(), break vs continue vs pass, loop-else clause, user-defined functions, *args and **kwargs, LEGB scope hierarchy, lambda functions, and datetime module operations.*

1. Control Flow & Loop Control Primitives

1. Conditional Branching & Truth Value Testing

Python evaluates conditional expressions using **truthy** and **falsy** semantics.

- **Falsy Constants:** `False`, `None`, `0`, `0.0`, `""` (empty string), `[]` (empty list), `()` (empty tuple), `{}` (empty dict), `set()`.

- **Truthy Values:** Any non-zero number or non-empty sequence/collection.

2. Comparative Analysis: break vs continue vs pass

Feature	`break`	`continue`	`pass`
Action	Terminates loop immediately	Aborts current iteration cycle	Null operation (NOP stub)
Target	First statement outside loop	Next loop cycle evaluation	Next sequential statement
Loop-else Effect	**Suppresses loop-else block!**	Loop-else executes normally	Zero effect
Use Case	Early exit upon finding item	Filtering invalid data items	Syntactic placeholder for empty code

```
# The for-else Search Idiom
for n in range(2, 10):
    for x in range(2, n):
        if n % x == 0:
            print(f"{n} equals {x} * {n//x}")
            break
    else:
        # Loop fell through without hitting a break!
        print(f"{n} is a prime number")
```

2. Function Architecture, Arguments & LEGB Scope Resolution

1. Function Argument Mechanics

Python supports 4 primary argument configurations:

1. **Positional Arguments:** Matched left-to-right by position.

2. **Keyword Arguments:** Matched explicitly by parameter name (`func(name="Alice", age=20)`).

3. **Default Arguments:** Assigned fallbacks (`def func(a, b=10)`). *Note: Mutable default argument trap (`def f(x=[])`) causes shared state; always use `None` as sentinel!*

4. **Variable-Length Arguments:**

- `*args` : Collects excess positional arguments into a **tuple**.
- `kwargs` **: Collects excess keyword arguments into a** dictionary**.**

2. The LEGB Name Resolution Protocol

When an unqualified variable is referenced inside a function, Python searches 4 concentric namespaces in strict order:

1. L (Local): **Names assigned inside the currently executing function.**

2. E (Enclosing): **Names in local scopes of enclosing functions from inner to outer (closures).**

3. G (Global): **Names assigned at the top-level of the current module file.**

4. B (Built-in):** Pre-assigned built-ins in `builtins` module (`open` , `len` , `range` , `print`).

```
# Date and Time Formatting Operations
from datetime import datetime, timedelta

now = datetime.now()
formatted = now.strftime("%d-%B-%Y %H:%M:%S")  # e.g., '19-September-2026 16:30:00'
parsed = datetime.strptime("19-09-2026", "%d-%m-%Y")
one_week_later = now + timedelta(days=7)       # Duration arithmetic
```